

STAT452 Midterm Tests

Comprehensive Solutions and R Guide

Applied Statistics for Engineers and Scientists II

Written Midterm and Practical Midterm

Included: full paper-based derivations, R verification, calculator checks, OLS diagnostics, nested-model testing, ridge and LASSO cross-validation, coefficient interpretation, and predictions.

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Guide Overview

This guide covers both papers in this folder:

- MidTerm_STAT452_23TT2.pdf - written midterm
- Midterm_Practical.pdf - R practical midterm

The companion script `midterm_solutions.R` reproduces all numerical results. The two linked datasets from the practical paper are saved as `seatpos.csv` and `newdata.csv`.

0.1 Quick answer sheet

Item	Main result
Written 1(a)	$y_{\text{hat}} = 1.819205 + 0.486520 x_1 + 0.558644 x_2$; R-squared = 0.984959
Written 1(b)	Add x_2 ; partial F = 117.42, p = 0.008409
Written 1(c)	Predicted score = 7.603669
Written 2(a)	$\log(y_{\text{hat}}) = 0.924456 + 0.241221 x$
Written 2(b)	Year-5 plug-in prediction = 8.419576 million
Written 3(a)	$\text{logit}(p_{\text{hat}}) = -3.449548 + 0.002294 x_1 + 0.777014 x_2 - 0.560031 x_3$
Written 3(b)	Predicted classes: 0, 0, 1, 0
Practical OLS	R-squared = 0.687; adjusted R-squared = 0.600; overall p = 1.31e-5
Practical ridge	With the fixed folds in the script, <code>lambda.min</code> = 36.521
Practical LASSO	<code>lambda.min</code> = 5.748; nonzero predictors are Age, Ht, and Leg

1 Part I - Written Midterm

1.1 Problem 1 - Multiple linear regression

1.1.1 Core idea

Multiple regression chooses the plane that minimizes the sum of squared vertical residuals. Here the plane uses both self-study time (x_1) and tutoring time (x_2) to predict the score y .

The model is

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \text{error}_i.$$

1.1.2 1(a) Estimate the coefficients and R-squared

Write the model in matrix form as $y = X \beta + \text{error}$, where the first column of X is a column of ones:

```
X = [1  5  1]      y = [ 5.0]
     [1  9  0]      [ 6.0]
     [1  7  7]      [ 9.5]
     [1  9  7]      [10.0]
     [1  3  8]      [ 7.5]
```

The least-squares estimator is

```
beta_hat = (X'X)^(-1) X'y.
```

For these data,

```
X'X = [  5   33   23]      X'y = [ 38.0]
       [ 33  245  141]      [258.0]
       [ 23  141  163]      [201.5]
```

Solving the normal equations gives

```
beta_0_hat = 1.819205
beta_1_hat = 0.486520
beta_2_hat = 0.558644
```

Therefore,

```
y_hat = 1.819205 + 0.486520 x1 + 0.558644 x2.
```

Interpretation, always holding the other predictor fixed:

- One additional hour of self-study is associated with an estimated 0.4865-point increase in score.
- One additional hour of tutoring is associated with an estimated 0.5586-point increase in score.
- The intercept is the fitted score at $x_1 = x_2 = 0$. It is mathematically needed, but the dataset contains no student near that combination, so it should not be given a strong practical interpretation.

The fitted residual sum of squares and total sum of squares are

```
SSE = sum((y_i - y_hat_i)^2) = 0.281267
SST = sum((y_i - y_bar)^2)   = 18.700000
```

Thus,

```
R-squared = 1 - SSE/SST
           = 1 - 0.281267/18.7
           = 0.984959.
```

About 98.50% of the observed score variation in these five students is explained by the fitted model. This is an in-sample description, not proof that study hours cause higher scores, and the sample is extremely small.

R verification:

```
d <- data.frame(
  x1 = c(5, 9, 7, 9, 3),
  x2 = c(1, 0, 7, 7, 8),
  y = c(5, 6, 9.5, 10, 7.5)
)
fit <- lm(y ~ x1 + x2, data = d)
coef(fit)
summary(fit)$r.squared
```

1.1.3 1(b) Compare $y \sim x_1$ with $y \sim x_1 + x_2$

The reduced model is nested inside the full model. The partial F test asks:

```
H0: beta_2 = 0 (x2 adds no linear information after x1)
H1: beta_2 != 0.
```

From the supplied ANOVA table,

```
F = 117.42, p = 0.008409.
```

At significance level 0.05, $p < 0.05$, so reject H_0 . The tutoring variable significantly reduces RSS after self-study time has already been included. Among the two proposed models, use the full model $y \sim x_1 + x_2$.

Do not compare the two models merely by noting that the full model has a smaller RSS: adding a predictor can never increase training RSS. The F test determines whether the reduction is large relative to the remaining error and degrees of freedom.

1.1.4 1(c) Prediction

At $x_1 = 5$ and $x_2 = 6$,

```
y_hat = 1.819205 + 0.486520(5) + 0.558644(6)
        = 7.603669.
```

The predicted math score is approximately **7.60**.

R verification:

```
predict(fit, newdata = data.frame(x1 = 5, x2 = 6))
```

1.1.5 Casio fx-580VN X verification

The calculator does not have a direct two-predictor regression menu. Use matrix mode to solve the three normal equations.

1. Open MENU -> Matrix.
2. Define MatA as the 3-by-3 matrix $X'X$ shown above.
3. Define MatB as the 3-by-1 matrix $X'Y$ shown above.

4. Calculate $\text{MatA}^{(-1)} * \text{MatB}$.
5. The display should give approximately $(1.819205, 0.486520, 0.558644)'$.
6. In ordinary calculation mode, substitute the five fitted values, compute the five squared residuals, and add them to verify $SSE = 0.281267$.

Menu wording can differ slightly with calculator language and firmware.

1.2 Problem 2 - Exponential regression

1.2.1 Core idea

Taking logarithms turns multiplicative growth into an ordinary straight-line regression.

The model is

$$y = \exp(\beta_0 + \beta_1 x + \text{error}).$$

Taking natural logarithms gives

$$z = \log(y) = \beta_0 + \beta_1 x + \text{error},$$

where $z = \log(y)$. The transformed observations are approximately

x	y	log(y)
1	2.98	1.091923
2	4.65	1.536867
3	5.00	1.609438
4	6.50	1.871802

1.2.2 2(a) Estimate the transformed linear model

For simple linear regression,

$$\begin{aligned}\beta_1_{\text{hat}} &= S_{xz}/S_{xx}, \\ \beta_0_{\text{hat}} &= \bar{z} - \beta_1_{\text{hat}} \bar{x}.\end{aligned}$$

Using the four transformed observations gives

$$\begin{aligned}\beta_1_{\text{hat}} &= 0.241221 \\ \beta_0_{\text{hat}} &= 0.924456.\end{aligned}$$

Hence,

$$\log(\hat{y}) = 0.924456 + 0.241221 x.$$

Equivalently, on the original scale,

```
y_hat = exp(0.924456) exp(0.241221)^x
      = 2.52050 (1.27280)^x.
```

The fitted population multiplies by about 1.2728 for each additional year, an estimated 27.28% year-to-year increase under this model.

R verification:

```
d <- data.frame(x = 1:4, y = c(2.98, 4.65, 5.00, 6.50))
fit_exp <- lm(log(y) ~ x, data = d)
coef(fit_exp)
```

1.2.3 2(b) Predict year 5

```
log(y_hat_5) = 0.924456 + 0.241221(5) = 2.130559,
y_hat_5 = exp(2.130559) = 8.419576.
```

The exam-style prediction is approximately **8.42 million people**.

A technical nuance: because the error is normal on the log scale, $\exp(\beta_0 + \beta_1 x)$ estimates the conditional median on the original scale. The conditional mean would include the lognormal correction $\exp(\sigma^2/2)$. The question asks for a prediction based on the fitted values, so the usual expected answer is the plug-in value 8.4196.

1.2.4 Casio fx-580VN X verification

1. Compute $\ln(y)$ for each of the four populations.
2. Open MENU $\rightarrow$ Statistics $\rightarrow$ A+BX.
3. Enter x in the first column and $\ln(y)$ in the second.
4. Open the regression calculation results and read the intercept a and slope b ; they should be approximately $a = 0.924456, b = 0.241221$.
5. Evaluate $\exp(a + 5b)$ to obtain approximately 8.419576.

1.3 Problem 3 - Logistic regression

1.3.1 Core idea

Logistic regression predicts a probability while keeping it between zero and one. Its linear predictor describes log-odds rather than the probability itself.

Let $p(x) = P(y = 1 \mid x_1, x_2, x_3)$. The estimated model is

```
log(p_hat/(1-p_hat))
= -3.449548 + 0.002294 x1 + 0.777014 x2 - 0.560031 x3.
```

Equivalently,


```
p_hat = exp(eta_hat) / (1 + exp(eta_hat)),
```

where η_{hat} is the right-hand side of the logit equation.

1.3.2 3(a) Interpret the coefficients

Each coefficient is a change in log-odds when that predictor rises by one unit, holding the other predictors fixed. Exponentiating gives the easier odds-ratio interpretation.

Term	Estimate	Odds ratio	Interpretation
x1	0.002294	1.002297	A one-unit increase multiplies the odds of class 1 by 1.0023, about a 0.23% increase.
x2	0.777014	2.174968	A one-unit increase multiplies the odds by about 2.175, a 117.5% increase.
x3	-0.560031	0.571191	A one-unit increase multiplies the odds by 0.571, about a 42.9% decrease.

For x_1 , one unit may be too small to communicate well. A 100-unit increase multiplies the odds by $\exp(100 * 0.002294)$, approximately 1.258.

The p-values for x_1 , x_2 , and x_3 are all below 0.05, so each is statistically significant at the 5% level in the fitted model. Significance does not by itself prove causation.

1.3.3 3(b) Classify the four observations

Use the standard cutoff 0.5: predict class 1 when $p_{\text{hat}} \geq 0.5$, equivalently when $\eta_{\text{hat}} \geq 0$.

Row	η_{hat}	p_{hat}	Predicted class
1	-1.058484	0.257599	0
2	-0.940450	0.280809	0
3	0.362391	0.589619	1
4	-0.684196	0.335326	0

Thus the predicted classes are **0, 0, 1, 0**.

R verification:

```
b <- c(-3.449548, 0.002294, 0.777014, -0.560031)
new <- data.frame(
  x1 = c(640, 600, 700, 620),
  x2 = c(3.35, 3.62, 3.56, 3.17),
  x3 = c(3, 3, 1, 2)
```

```

)
eta <- drop(cbind(1, as.matrix(new)) %*% b)
p <- plogis(eta)
cbind(new, eta, p, class = as.integer(p >= 0.5))

```

1.3.4 Casio fx-580VN X verification

For each row, use ordinary calculation mode:

1. Compute $\eta = -3.449548 + 0.002294 \cdot x_1 + 0.777014 \cdot x_2 - 0.560031 \cdot x_3$.
2. Compute $1/(1 + \exp(-\eta))$.
3. Compare the result with 0.5.

For the first row the display should give η about -1.05848 and probability about 0.25760, so the class is 0.

1.4 Problem 4 - Ridge estimator when the intercept is penalized

1.4.1 What we are trying to show

We want to minimize

$$L = \sum ((y_k - \beta_0 - \beta_1 x_k)^2) + \lambda (\beta_0^2 + \beta_1^2)$$

and derive the two formulas printed in the question.

The strategy is to differentiate with respect to both coefficients, solve the first normal equation for β_0 , and substitute it into the second.

1.4.2 Step 1: Differentiate with respect to β_0

Set the derivative equal to zero:

$$-2 \sum (y_k - \beta_0 - \beta_1 x_k) + 2 \lambda \beta_0 = 0.$$

Using $\sum (y_k) = n \bar{y}$ and $\sum (x_k) = n \bar{x}$ gives

$$(n + \lambda) \beta_0 + n \bar{x} \beta_1 = n \bar{y}.$$

Therefore,

$$\hat{\beta}_0 = n / (n + \lambda) (\bar{y} - \hat{\beta}_1 \bar{x}).$$

This differs from ordinary least squares because the question penalizes the intercept. Without an intercept penalty, the familiar result would be $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$.

1.4.3 Step 2: Differentiate with respect to β_1

Set the second derivative equation to zero:

$$-2 \sum (x_k (y_k - \beta_0 - \beta_1 x_k)) + 2 \lambda \beta_1 = 0,$$

so

$$\sum (x_k y_k) = n \bar{x} \beta_0 + \beta_1 [\sum (x_k^2) + \lambda].$$

Substitute the expression for β_0 :

$$\begin{aligned} \sum (x_k y_k) &= n \bar{x} * n / (n + \lambda) (\bar{y} - \beta_1 \bar{x}) \\ &\quad + \beta_1 [\sum (x_k^2) + \lambda]. \end{aligned}$$

Collect the terms involving β_1 :

$$\begin{aligned} \beta_1 * [\sum (x_k^2) + \lambda - n^2 \bar{x}^2 / (n + \lambda)] \\ = \sum (x_k y_k) - n^2 \bar{x} \bar{y} / (n + \lambda). \end{aligned}$$

1.4.4 Step 3: Convert to centered sums

Use

$$\begin{aligned} \sum (x_k y_k) &= S_{xy} + n \bar{x} \bar{y}, \\ \sum (x_k^2) &= S_{xx} + n \bar{x}^2. \end{aligned}$$

The numerator becomes

$$\begin{aligned} S_{xy} + n \bar{x} \bar{y} - n^2 \bar{x} \bar{y} / (n + \lambda) \\ = S_{xy} + [n \lambda / (n + \lambda)] \bar{x} \bar{y}. \end{aligned}$$

The denominator becomes

$$\begin{aligned} S_{xx} + n \bar{x}^2 + \lambda - n^2 \bar{x}^2 / (n + \lambda) \\ = S_{xx} + \lambda + [n \lambda / (n + \lambda)] \bar{x}^2 \\ = S_{xx} + \lambda [1 + n \bar{x}^2 / (n + \lambda)]. \end{aligned}$$

Hence,

$$\begin{aligned} \hat{\beta}_1 &= \\ &\frac{\{S_{xy} + [n \lambda / (n + \lambda)] \bar{x} \bar{y}\}}{\{S_{xx} + \lambda [1 + n \bar{x}^2 / (n + \lambda)]\}}, \end{aligned}$$

and

$$\hat{\beta}_0 = n / (n + \lambda) (\bar{y} - \hat{\beta}_1 \bar{x}),$$

which are exactly the required formulas.

1.4.5 Important distinction from `glmnet`

The theoretical problem explicitly adds $\lambda * \beta_0^2$, so it penalizes the intercept. By default, `glmnet` does **not** penalize the intercept. Do not expect the practical ridge intercept to follow this written-

problem formula.

There is no useful direct calculator shortcut for this symbolic derivation. The exam expects the normal-equation argument above.

2 Part II - Practical Midterm

2.1 Reproducible setup

The training file has 38 rows and these columns:

```
Age Weight HtShoes Ht Seated Arm Thigh Leg hipcenter
```

Run the complete analysis from the repository root with:

```
Rscript R_training/midterm/midterm_solutions.R
```

Or, in RStudio:

```
source("R_training/midterm/midterm_solutions.R")
```

Only `glmnet` is required beyond base R.

2.2 Practical (a) - Full OLS model and residual analysis

The estimated model is

```
hipcenter_hat = 436.4321
+ 0.7757 Age
+ 0.0263 Weight
- 2.6924 HtShoes
+ 0.6013 Ht
+ 0.5338 Seated
- 1.3281 Arm
- 1.1431 Thigh
- 6.4390 Leg.
```

Every slope is interpreted while all seven other variables are held fixed. For example, one additional year of age is associated with a 0.776 mm increase in the fitted `hipcenter` coordinate at the same weight and body dimensions. A one-centimeter increase in lower-leg length is associated with a 6.439 mm decrease in the fitted coordinate, holding the remaining measurements fixed.

Be cautious: holding `Ht` fixed while changing `HtShoes`, or vice versa, is nearly an impossible comparison. The coefficient table reflects severe collinearity.

2.2.1 Model fit

```
Residual standard error = 37.7 mm on 29 df
R-squared                  = 0.687
Adjusted R-squared        = 0.600
```

```
Overall F(8,29)      = 7.94
Overall p-value      = 1.31e-5
```

The model explains about 68.7% of the observed training variation, or 60.0% after the adjusted-R-squared penalty for eight predictors.

2.2.2 Individual coefficients

None of the eight slopes has an individual p-value below 0.05. This does not contradict the highly significant overall F test. The predictors can be useful collectively while their individual standard errors are inflated by multicollinearity.

The clearest evidence is:

```
VIF(Ht)      = 333.14
VIF(HtShoes) = 307.43
standardized-predictor condition number = 59.77
```

Ht and HtShoes measure almost the same physical quantity. Their OLS coefficients can therefore become large, unstable, and even change sign while their combined fitted contribution remains useful. This is exactly the setting where ridge or LASSO can help.

2.2.3 Residual checks

1. **Centering and shape.** The residual median is -3.68 mm and the residual range is about -73.83 to 62.34 mm. OLS residuals sum to zero by construction.
2. **Normality.** Shapiro-Wilk gives $W = 0.9715$, $p = 0.434$. At 5%, there is no evidence against normal residuals. This is not proof of exact normality.
3. **Constant variance.** The base-R Breusch-Pagan calculation gives $LM = 14.037$, $df = 8$, $p = 0.0808$. At 5%, there is no statistically significant evidence of heteroscedasticity, although the p-value is close enough to justify checking the scale-location plot.
4. **Influence.** Row 31 has Cook's distance 0.695 and leverage 0.560. It exceeds both rough thresholds $4/n = 0.105$ and $2k/n = 0.474$, so it deserves data verification and a sensitivity refit. Rows 23 and 35 also have Cook's distances above $4/n$.
5. **Linearity.** Inspect the residuals-versus-fitted plot. A random cloud around zero supports the linear mean structure; curvature suggests transformations or nonlinear terms.

Interactive diagnostic plots:

```
par(mfrow = c(2, 2))
plot(full_ols)
par(mfrow = c(1, 1))
```

Do not delete an influential row only because Cook's distance is large. First check for a recording error, understand why it is unusual, and report whether the scientific conclusion changes when it is included or excluded.

2.3 Practical (b) - Full model versus noise-only model

The hypotheses are

```
H0: beta_Age = beta_Weight = ... = beta_Leg = 0
H1: at least one slope is nonzero.
```

The nested-model ANOVA gives

```
RSS(null) = 131639
RSS(full) = 41262
F(8,29) = 7.94
p = 1.31e-5.
```

Because $p < 0.05$, reject H_0 . The full set of anthropometric predictors gives a statistically significant improvement over an intercept-only model.

R code:

```
null_ols <- lm(hipcenter ~ 1, data = seatpos)
full_ols <- lm(hipcenter ~ ., data = seatpos)
anova(null_ols, full_ols)
```

2.4 Practical (c) - Ridge regression with cross-validation

Ridge minimizes squared error plus an L2 penalty:

```
SSE + lambda * sum(beta_j^2).
```

It shrinks correlated coefficients toward zero but normally keeps all of them nonzero. The script uses:

- `alpha = 0` for ridge;
- the exam grid `seq(1e-3, 50, by = 1e-3)`;
- 10-fold CV;
- `set.seed(452)` and explicit fold IDs;
- `type.measure = "mse"`;
- `standardize = TRUE`, the `glmnet` default.

With those fixed folds:

```
lambda.min = 36.521
lambda.1se = 50.000
```

`lambda.min` minimizes estimated cross-validation MSE. `lambda.1se` is the largest penalty whose error is within one standard error of the minimum and is a reasonable more-regularized alternative. Because `lambda.1se` hits the upper edge 50, a real analysis using the 1-SE rule should expand the grid above 50.

Ridge coefficients at `lambda.min`:

Term	Ridge estimate
Intercept	399.992503
Age	0.473581
Weight	-0.106186
HtShoes	-0.738541
Ht	-0.737116
Seated	-1.203317
Arm	-1.354624
Thigh	-1.304223
Leg	-3.109081

Compared with OLS, the unstable height coefficients and other slopes have been pulled toward a more balanced set of values. Ridge is especially appropriate when prediction is more important than choosing a small subset of variables.

`glmnet` standardizes predictors internally when fitting, does not penalize the intercept, and transforms the displayed coefficients back to the original units.

2.5 Practical (d) - Ridge predictions

Using `lambda.min = 36.521`:

New row	Ridge prediction (mm)
1	-139.711
2	-112.954
3	-219.990
4	-149.057
5	-228.157
6	-102.596

R code:

```
predict(ridge_cv, newx = X_new, s = "lambda.min")
```

Use `model.matrix` for both training and new data so that columns and encoding match exactly.

2.6 Practical (e) - LASSO and comparison with OLS

LASSO uses an L1 penalty:

```
SSE + lambda * sum(abs(beta_j)).
```

The corners of the L1 constraint allow some estimated slopes to be exactly zero, so LASSO performs shrinkage and variable selection at the same time.

With the same fixed folds:

```
lambda.min = 5.748
lambda.1se = 20.288
```

At `lambda.min`, the nonzero predictors are **Age, Ht, and Leg**.

Term	Full OLS	Ridge	LASSO
Intercept	436.432128	399.992503	397.696648
Age	0.775716	0.473581	0.221804
Weight	0.026313	-0.106186	0
HtShoes	-2.692408	-0.738541	0
Ht	0.601345	-0.737116	-2.200605
Seated	0.533752	-1.203317	0
Arm	-1.328069	-1.354624	0
Thigh	-1.143119	-1.304223	0
Leg	-6.439046	-3.109081	-5.468787

The LASSO model is therefore

```
hipcenter_hat = 397.696648
+ 0.221804 Age
- 2.200605 Ht
- 5.468787 Leg.
```

If ordinary least squares is refitted using only these three selected variables, the estimates are approximately

```
Intercept = 452.198
Age       = 0.581
Ht        = -2.325
Leg       = -6.739
```

The post-LASSO OLS values are less shrunken because that refit has no penalty. It is useful as a comparison, but it is not the LASSO estimator and naive OLS p-values after data-driven variable selection should not be treated as ordinary pre-specified inference.

Selection among highly correlated measurements can be unstable: another fold split may choose `HtShoes` instead of `Ht`, or slightly change the optimal `lambda`. Report the seed and folds, and focus on predictive validation rather than treating one selected set as a scientific law.

2.7 Practical (f) - Ridge versus LASSO predictions

Row	Ridge	LASSO	Ridge - LASSO
1	-139.711	-130.368	-9.344
2	-112.954	-112.470	-0.485
3	-219.990	-218.053	-1.937
4	-149.057	-151.148	2.091
5	-228.157	-229.846	1.689
6	-102.596	-100.072	-2.524

The two regularized models give similar predictions for five of the six rows. Row 1 differs by about 9.34 mm. Without observed `hipcenter` values for these six rows, the table compares predictions but cannot determine which model predicts better. That requires held-out outcomes or repeated cross-validation error.

3 Practical Exam Workflow

Use this order under time pressure:

1. Read the CSV files and immediately check `dim`, `names`, `str`, and missing values.
2. Fit `lm(hipcenter ~ ., data = seatpos)` and save the model object.
3. Interpret slopes with the phrase "holding all other predictors fixed."
4. Report R-squared, adjusted R-squared, residual standard error, and the overall F test.
5. Inspect residual plots, normality, constant variance, leverage, Cook's distance, and multicollinearity.
6. Compare the null and full OLS models with `anova(null_ols, full_ols)`.
7. Construct `X` with `model.matrix` and remove only its intercept column.
8. Fix the random seed and fold IDs before fitting both ridge and LASSO.
9. Use the same folds for a fair comparison.
10. State whether `lambda.min` or `lambda.1se` is used for coefficients and predictions.
11. Build `X_new` with the same column construction as `X`.
12. Compare predictions numerically; do not claim a winner without actual test outcomes.

3.1 Common mistakes

- Interpreting a logistic coefficient as a direct change in probability.

- Forgetting to back-transform the exponential model with `exp()`.
- Claiming that high R-squared proves causation.
- Saying the full OLS model is useless because no individual slope is significant, despite a significant overall F test and severe collinearity.
- Passing a data frame directly to `glmnet`; it needs a numeric matrix.
- Leaving the intercept column in `x` when `glmnet` already fits an intercept.
- Using different CV folds for ridge and LASSO when comparing them.
- Reporting an optimal lambda without a seed or fold definition.
- Treating the LASSO-selected variables as stable when predictors are highly correlated.
- Comparing ridge and LASSO predictions without observed outcomes and declaring one model more accurate.

3.2 One-line summary

The written test checks whether you can translate models into estimators and interpretations; the practical test shows why regularization is valuable when a small dataset contains strongly correlated predictors.